

Diversity dependence in RevBayes

A tree-only likelihood benchmark against DDD, followed by a simulation pilot

100

simulated trees

481

checks in the Rev executable

5.2×10^{-12}

largest Rev-versus-DDD difference

Result: the implementations agree numerically under matched rate laws, sampling and conditioning. Parameter estimates still vary substantially among simulated trees.

GENERATE

Two rate models; crown ages 3 and 8; 25 trees per scenario. $\lambda_0 = 0.8$, $\mu = 0.2$, $K = 15$.

COMPARE

Evaluate identical parameter points and fitted points. Check the actual Rev function, not just a standalone prototype.

ESTIMATE

Fit λ_0 only, holding μ and K at their generating values. Retain all trees and all boundary estimates.

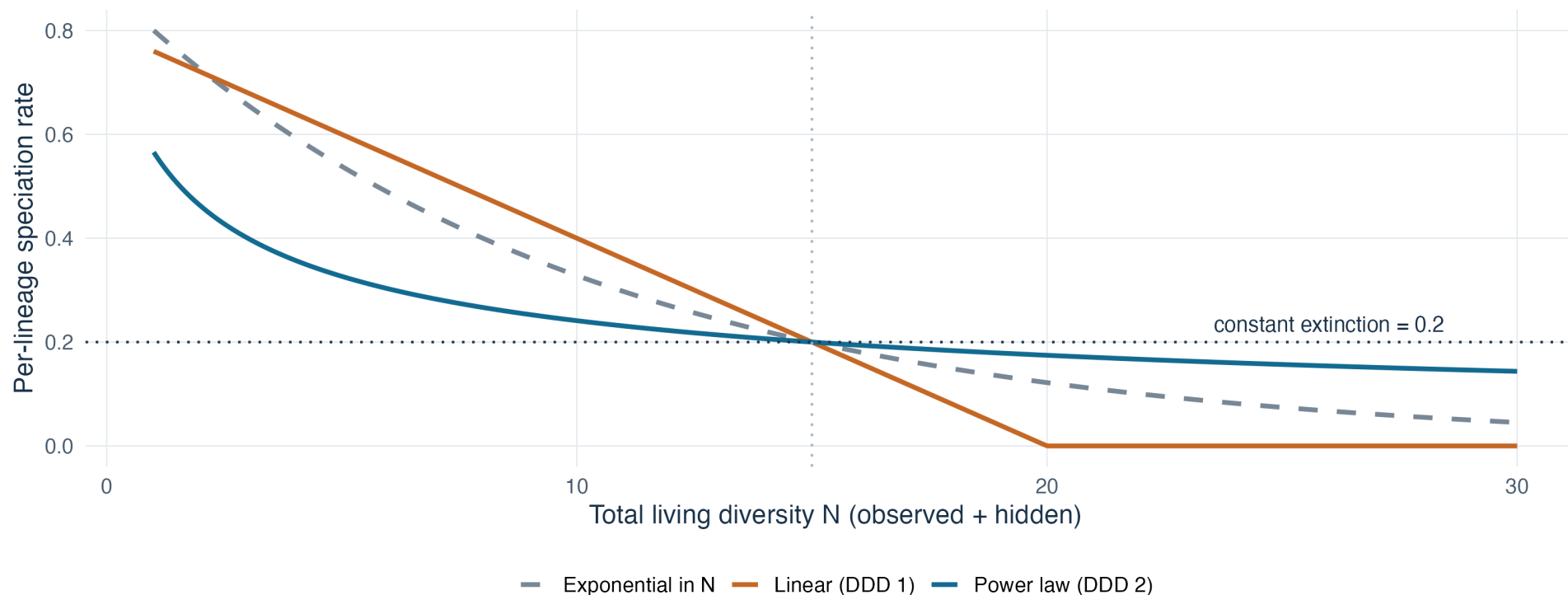
Scope: a numerical benchmark and a small estimation experiment. This does not establish joint parameter recovery or validate fossil-data MCMC.

Speciation responds to total living diversity

$N(t)$ includes both reconstructed lineages and lineages with no living sampled descendants.

Different rate laws, shared dependence on total diversity

Simulation models: $\lambda_{00} = 0.8$, $\mu = 0.2$, $K = 15$. Grey curve: $\alpha = \log(4)/14$; shown for contrast only.



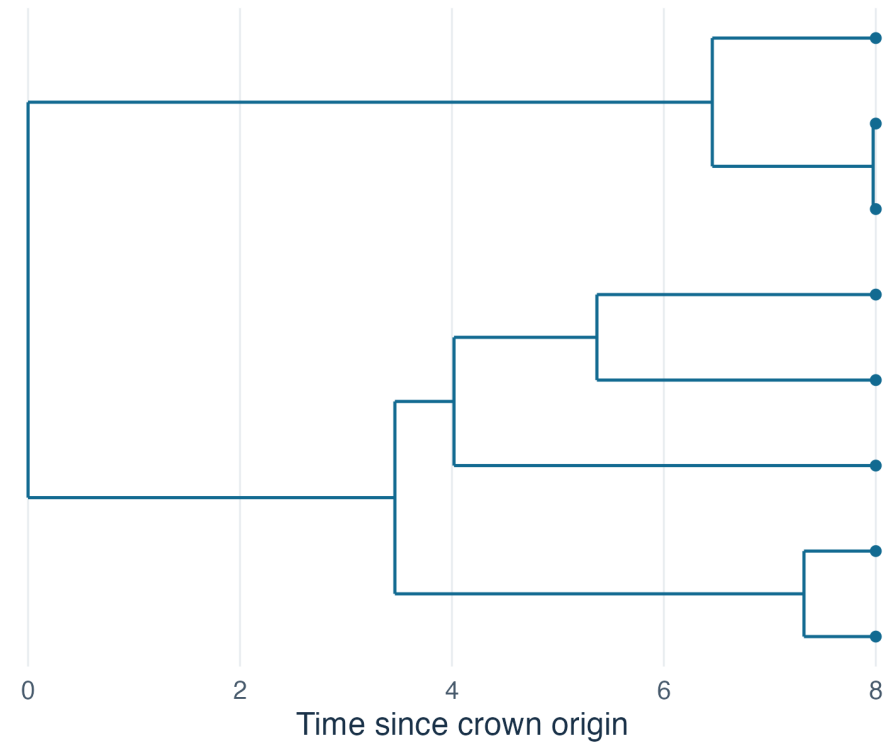
The simulations use DDD's linear model and its power-law model. The latter is called "exponential" in DDD documentation, but it is not $\exp(-\alpha N)$. Both simulated curves satisfy $\lambda(K) = \mu$; K is not a hard cap on diversity.

The observed tree contains only part of the history

Extinct lineages affect speciation rates even when they leave no tip in the reconstructed tree.

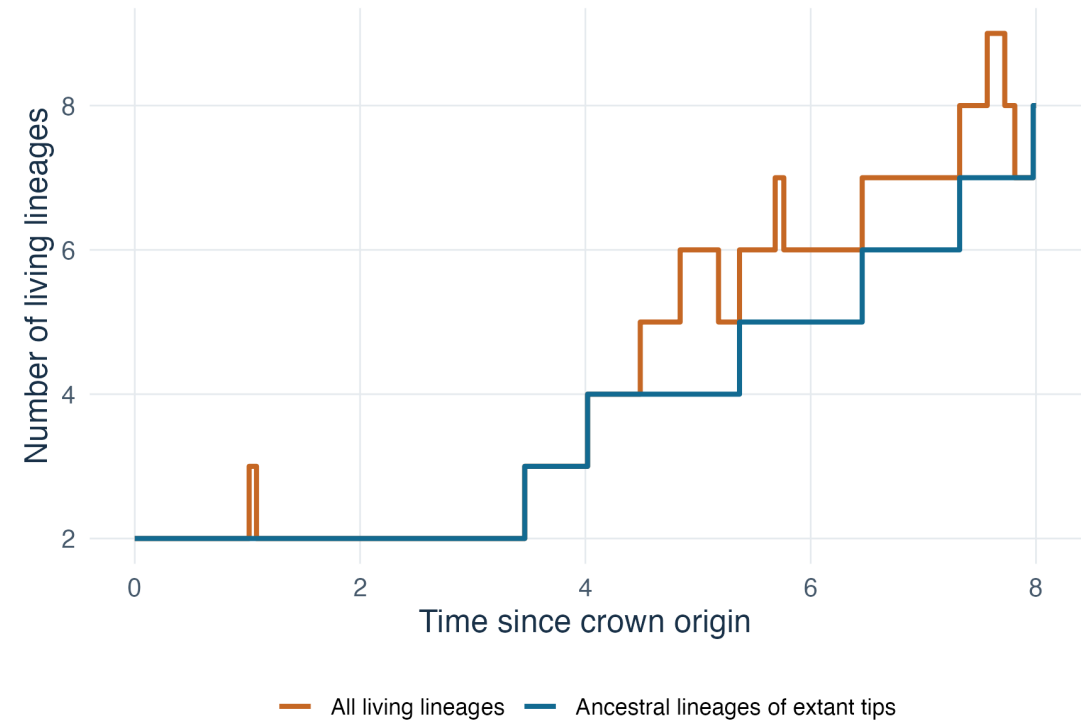
Observed tree: 8 living species

Illustrative history: DDDpower_age8_rep06



The likelihood integrates over hidden history

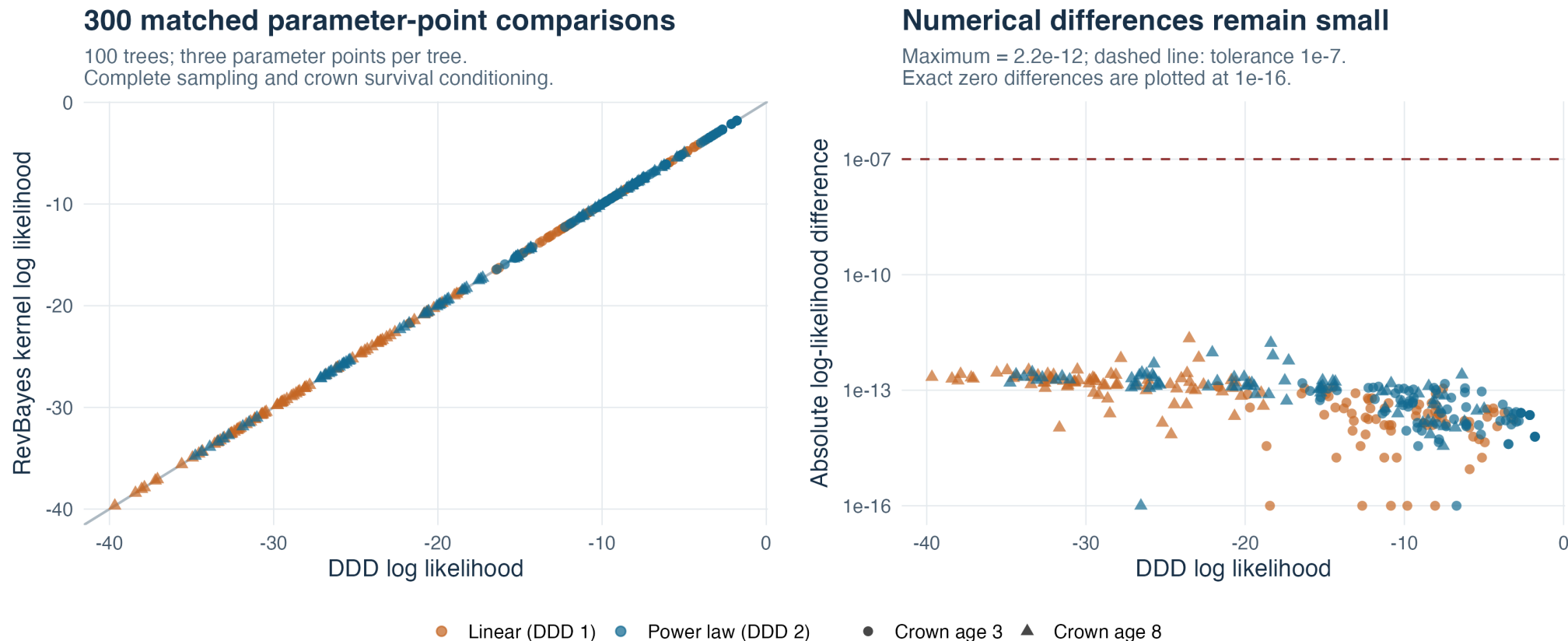
13 species originated; 5 became extinct.



This history contains 13 species in total, including 5 extinct species and 8 living tips. The likelihood receives only the extant tree. Illustration selected as the tree closest to the median tip count in the power-law, age-8 group; all 100 trees remain in the analyses.

The same model gives the same likelihood

No arbitrary normalization offset was fitted or subtracted.



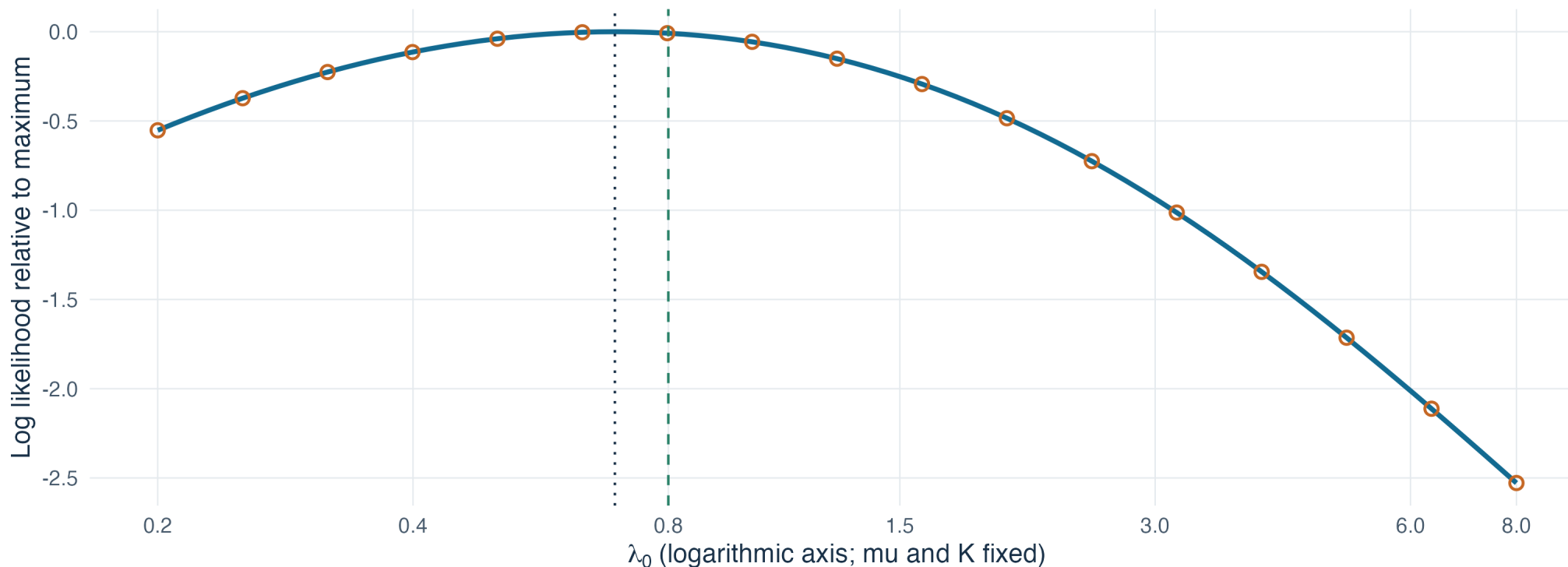
Three parameter points per tree give a maximum absolute difference of 2.2×10^{-12} . The full Rev check includes these 300 points, 100 fitted points and an 81-point profile: 481 checks, maximum difference 5.2×10^{-12} . All use crown survival, complete sampling and DDD's phylogeny-density convention.

Independent fits trace the same likelihood profile

Common scalar search and parameter bounds; extinction and K held fixed.

The two likelihood profiles overlap

Blue line: RevBayes; orange circles: DDD. Generating $\lambda_0 = 0.8$ (green); fitted $\lambda_0 = 0.692$ (grey).



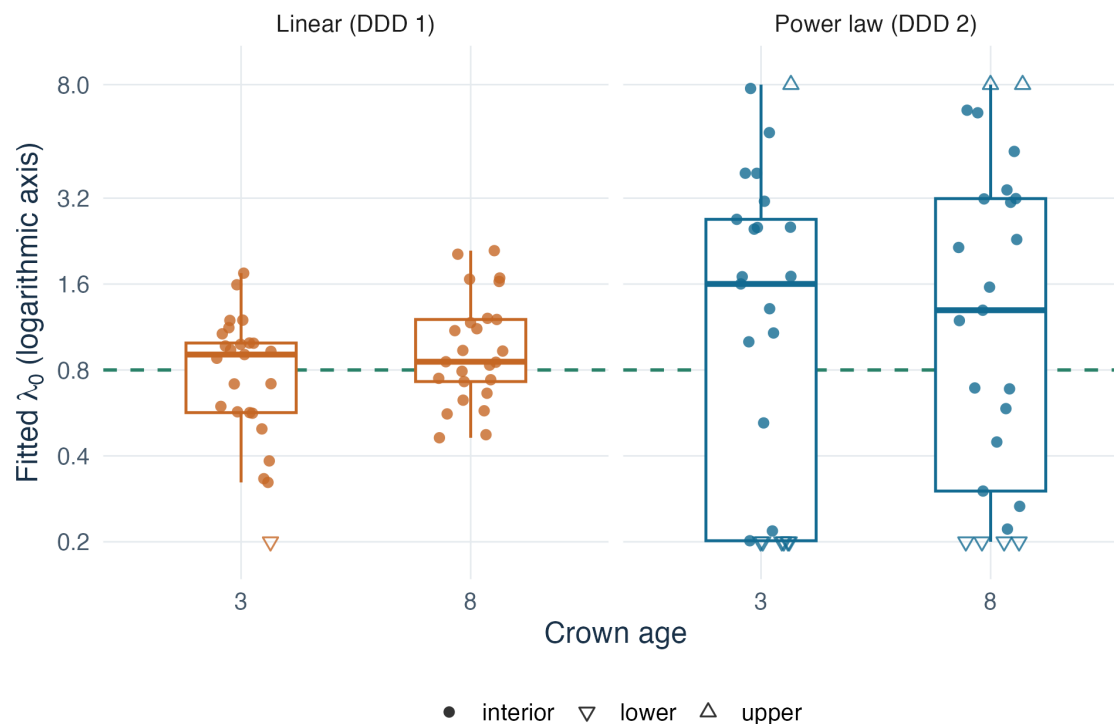
For this illustrative tree, $\hat{\lambda}_0 = 0.692$, compared with the generating value 0.8. Across all 100 fits, the largest fitted log-likelihood difference is $7.9\text{e-}12$; the largest $\hat{\lambda}_0$ difference is $2.1\text{e-}06$. Fitting calls either DDD or the shared C++ kernel; final points are also checked in Rev.

Estimates vary even when the software agrees

A 25-replicate-per-scenario pilot, with all accepted trees and search-bound estimates retained.

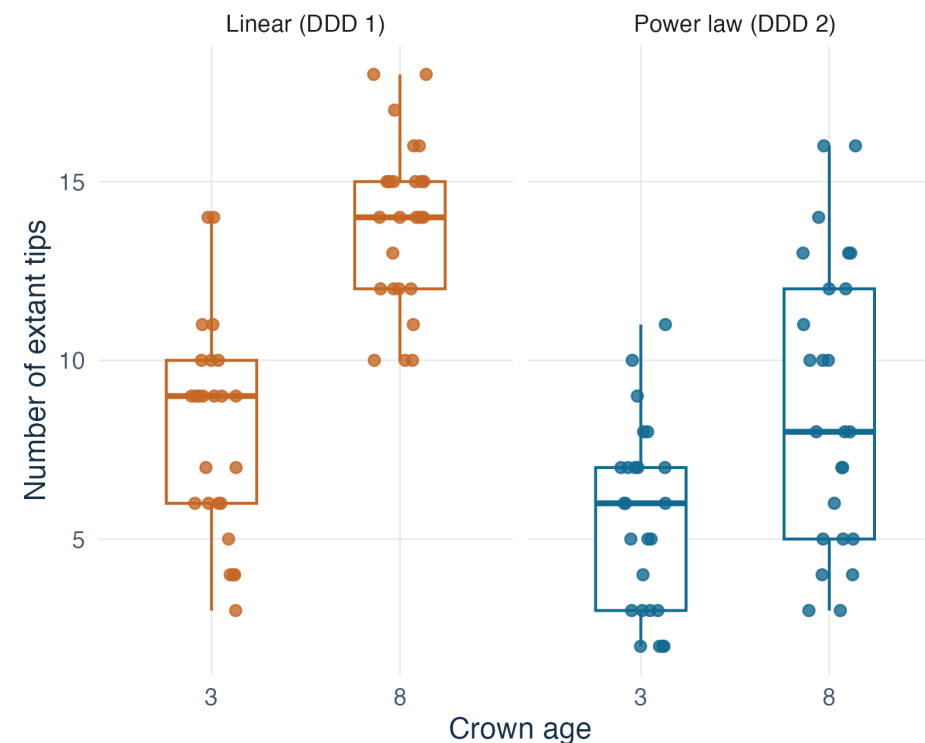
Matching software does not remove sampling variation

25 trees per group. Green: generating value 0.8. Triangles: search-bound estimates.



Information varies with realized tree size

All accepted simulated trees retained; no filtering by tip count.



11 fits reached the lower bound (0.20001) and 3 reached the upper bound (8). Triangles identify these restricted-search results. They are not evidence that an unconstrained optimum lies at exactly that value. With μ and K known, this is an easier inference problem than estimating all three parameters.

Pilot estimates remain variable across scenarios

Generating $\lambda_0 = 0.8$. Time units are arbitrary; rates are per lineage per unit time.

Rate model	Crown age	Trees	Extant tips median [range]	λ_0 median [IQR]	Bound hits lower / upper
Linear	3	25	9 [3-14]	0.906 [0.567-0.994]	1 / 0
Linear	8	25	14 [10-18]	0.855 [0.729-1.202]	0 / 0
Power law	3	25	6 [2-11]	1.602 [0.202-2.698]	6 / 1
Power law	8	25	8 [3-16]	1.296 [0.301-3.189]	4 / 2

Interpretation

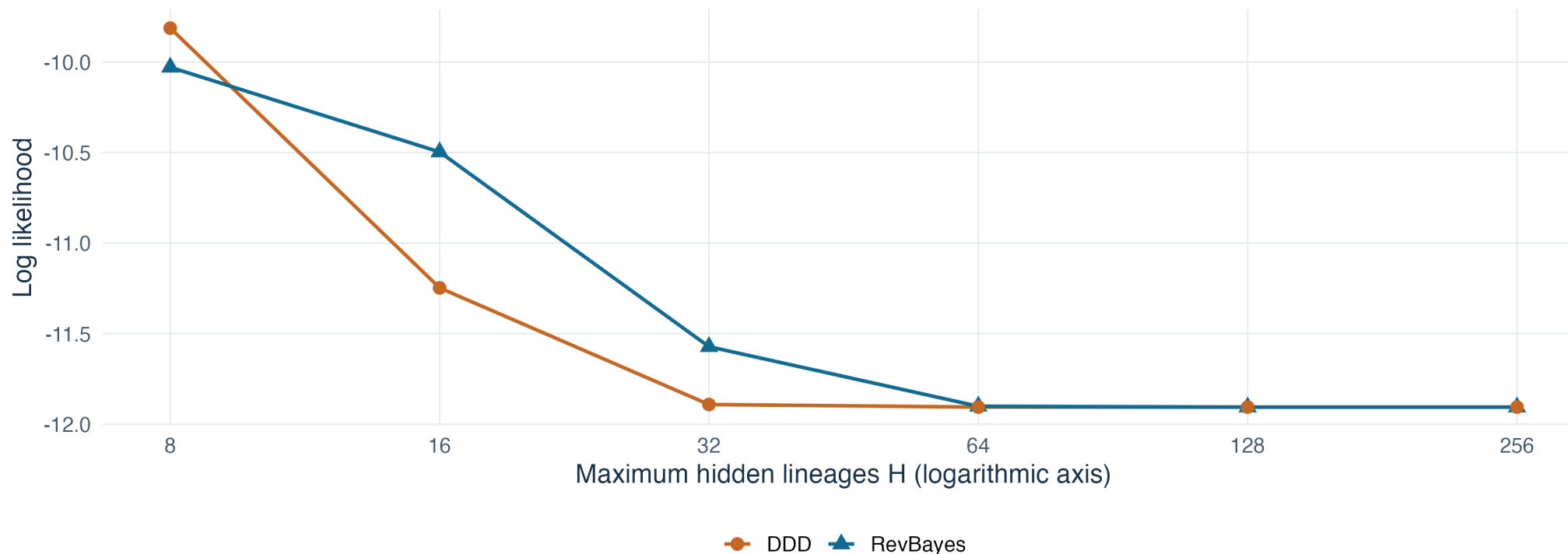
The power-law runs show wide variation and search-bound estimates at both ages. This pilot does not support a claim of reliable joint recovery, or a general claim that older trees solve the estimation problem. IQRs describe variation among simulated point estimates; they are not confidence intervals.

Hidden-lineage truncation must be checked

Solver tolerance controls propagation accuracy; it does not control the hidden-state cutoff error.

Converge the hidden-lineage cutoff before interpreting results

Stress case: power law, $\lambda_0 = 1.2$, $\mu = 0.8$, $K = 30$; crown age 10; internal ages 10, 6, 3.



This earlier stress case deliberately uses insufficient cutoffs. Rev kills probability escaping the upper boundary; DDD's matrix backend modifies its top diagonal. Their low-cutoff values need not match. Rev's $H=256$ to $H=512$ change is $4.4e-14$. In the 100 new fits, $H=128$ to $H=256$ changes log likelihood by at most $2.1e-13$.

Run the bundled example in RevBayes

A self-contained six-tip teaching example; its $K = 8$ differs from the simulation study's $K = 15$.

```
tree = readTrees("examples/DDD/data/simulated.tre")[1]
lnL := fnDiversityDependentLogLikelihood(
  tree, lambda0=0.8, mu=0.2, K=8,
  rateModel="DDDpower", start="crown",
  condition="survival", density="DDDphylogeny",
  maxHiddenLineages=128, numericalTolerance=1e-13)
print(lnL)
```

Expected log likelihood: **−7.9263828301537**. The complete script also checks $H=256$, evaluates the other two rate laws and writes a λ_0 likelihood profile.

From the repository root:

```
.local-build/revbayes-build/rb examples/DDD/likelihood.Rev
```

Independent R reference: `Rscript examples/DDD/compare_in_R.R`

This is a function in the development build. A deterministic `lnL` variable alone does not add a likelihood factor to an MCMC model.

Model and likelihood conventions

Per-lineage speciation depends on total living diversity N ; per-lineage extinction μ is constant.

RATE LAWS

Linear (DDD model 1)

$$\lambda_N = \max[0, \lambda_0 - (\lambda_0 - \mu)N/K]$$

Power law (DDD model 2)

$$\lambda_N = \lambda_0(N + 1)^{-a}$$

$$a = \log(\lambda_0/\mu) / \log(K + 1)$$

Original exponential form

$$\lambda_N = \lambda_0 \exp[-\alpha(N - 1)]$$

Only the first two forms are simulated here. K denotes the diversity at which λ equals μ ; it is not the maximum possible number of species.

HIDDEN-LINEAGE CALCULATION

Let k be reconstructed lineages and h hidden living lineages, so $N = k + h$.

$$A[h, h] = -(k + h)(\lambda_{k+h} + \mu)$$

$$A[h + 1, h] = (h + 2k)\lambda_{k+h}$$

$$A[h - 1, h] = h\mu$$

Propagate the weighted tree-embedding vector between nodes. At an observed branching, multiply by λ_{k+h} and increase k . At the present, select $h = 0$ for complete extant sampling.

A crown start has $k = 2$ and omits the root birth factor. Condition on both crown sides surviving in the coupled diversity process. A is an embedding operator, not the unconditional diversity-count generator.

Methods, limitations and reproducibility

Saved trees, seeds, numerical results, executable checks and plotting code accompany the report.

Simulation

DDD::dd_sim; models 1 and 2; $\lambda_0 = 0.8$, $\mu = 0.2$, $K = 15$; crown ages 3 and 8; 25 replicates each. DDD retains histories in which both crown sides survive. No additional filtering by tree size. Seeds: 20261001 + model $\times$ 10000 + age $\times$ 100 + replicate.

Likelihood and fitting

Complete extant sampling; crown survival condition; btorph=1. H=128, uniformization tolerance 10^{-13} . Three parameter points per tree: linear $\lambda_0 = 0.4, 0.6, 0.8$; power $\lambda_0 = 0.6, 0.8, 1.2$. Fit λ_0 within [0.20001, 8], with μ and K fixed, using a 13-point logarithmic scan, local scalar optimization and explicit endpoint checks.

Limits

Only extant trees and one fitted parameter are examined here. No fossils, character data, incomplete sampling, joint $\lambda_0/\mu/K$ recovery, confidence-interval coverage or fossil-model MCMC is validated. Twenty-five trees per group constitute a pilot; bound hits and finite search ranges constrain interpretation.

Reproduce

Rscript validation/DDD_presentation/simulate_and_compare.R; run results/verify_in_Rev.Rev with rb; Rscript validation/DDD_presentation/make_figures.R; python3 validation/DDD_presentation/build_report.py. See the accompanying README for commands and dependencies.

References

Etienne et al. (2012), Proc. R. Soc. B 279:1300-1309. doi:[10.1098/rspb.2011.1439](https://doi.org/10.1098/rspb.2011.1439)
 DDD package and source: rsetienne.github.io/DDD. Installed reference: DDD 5.2.5; session details saved in results/session.txt.

Tested binary SHA-256: bcda2e600fd567d937aa... | Full hashes and numerical checks: results/manifest.json